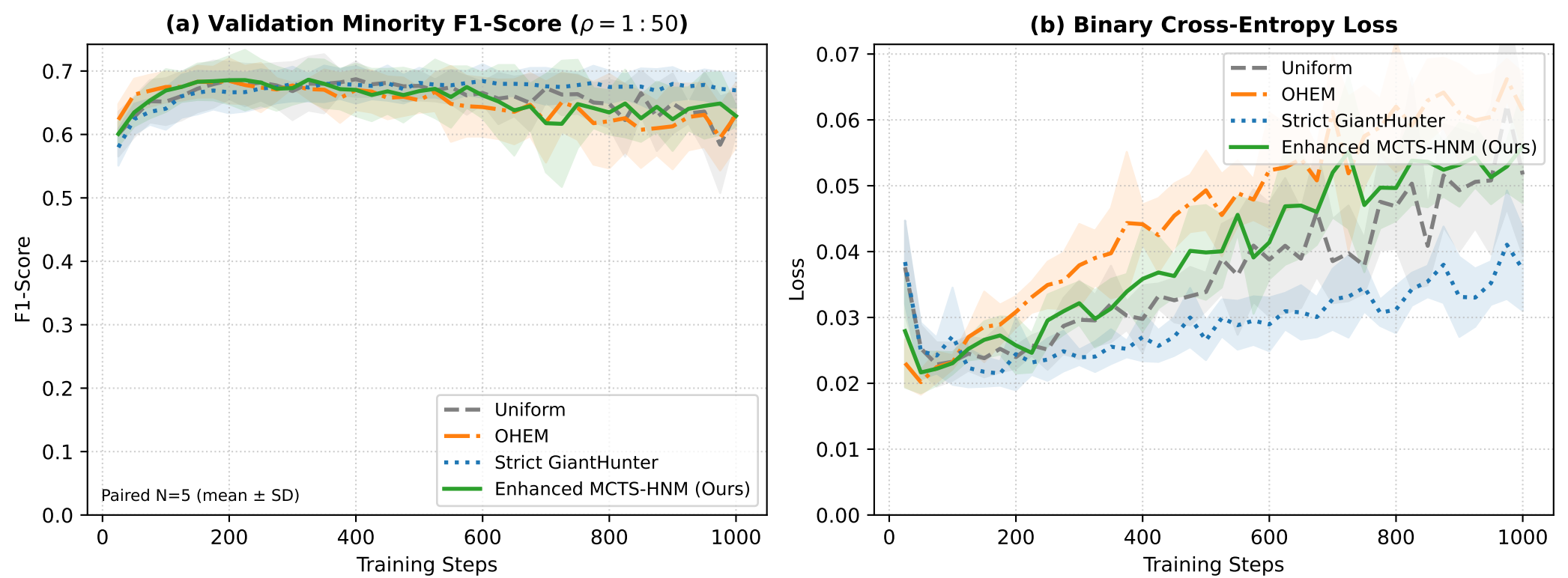
(a) Validation Minority F1-Score ($\rho = 1:50$)
F1-Score
Training Steps
Paired N=5 (mean ± SD)
Uniform
OHEM
Strict GiantHunter
Enhanced MCTS-HNM (Ours)
(b) Binary Cross-Entropy Loss
Loss
Training Steps
Uniform
OHEM
Strict GiantHunter
Enhanced MCTS-HNM (Ours)